Freight Rate Prediction

Data split, validation approach and December forecast | Derick William de Moraes Frias

Summary. The model is scored on November and December 2025, after all the labeled data, so every evaluation in this project trains on earlier months and tests on later ones. Model comparison used expanding-window cross-validation over January to September; October was held out and opened once, after the final model was frozen. The final model (a Ridge regression base with a time trend, corrected by LightGBM) reaches a cross-validated MAPE of **4.15% ± 0.32** against **10.31%** for the baseline, and **4.54%** on the October test (**1.56%** on loads with plausible labels).

1. Data split

The labeled data (`train-test.csv`) covers 48,000 loads from January to October 2025. The 12,000 loads to predict cover November and December. A random split would let the model learn from loads in the same weeks as the ones it is tested on, sharing the same market conditions, and would overstate accuracy. The data was therefore split **by month**, so the model is always evaluated on months later than the ones it learned from, exactly as in the real task.

Set	Months	Loads	Use
Train	Jan - Aug	38,477	Fit models
Validation	Sep	4,670	Compare models and decisions (last cross-validation fold)
Test	Oct	4,853	One final evaluation, after the model was frozen

Test-set rule. October was never explored, plotted or used for any decision. It was opened once, at the end. For the submission, the frozen configuration was refit on all of January to October.

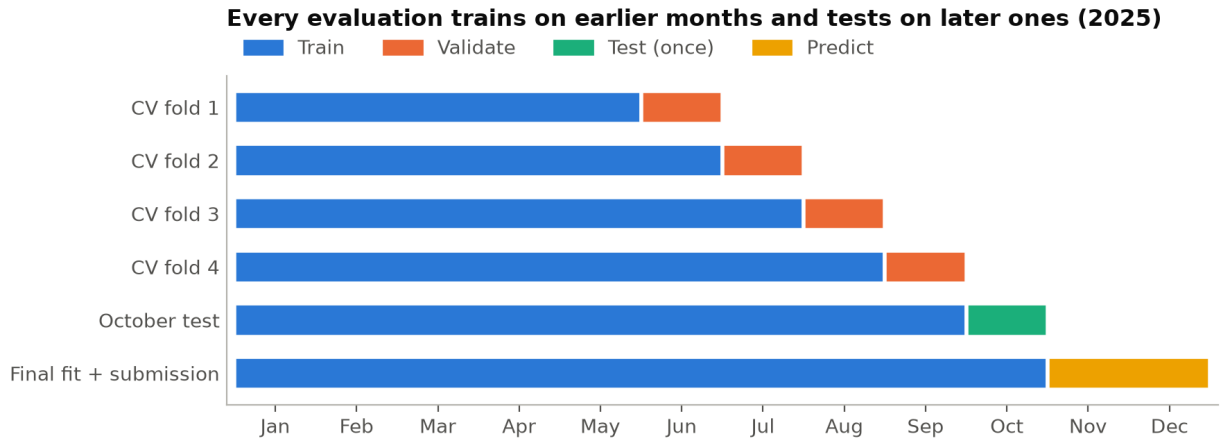


Figure 1. Months used for training, validation, testing and prediction in each evaluation round.

2. Validation approach

2.1 Expanding-window cross-validation

A single validation month gives a single, noisy score. Models were compared on four folds instead: each fold trains on every month before its validation month (June, July, August, September). Fold 4 is the train / validation split above. Models are ranked on the **mean** error across folds; the **standard deviation** shows whether a model is consistently good or only good in one market regime (June is the price peak, August and September are lows).

2.2 Metric

Rates range from about $150 to $8,000, so the primary metric is **MAPE** (mean absolute percentage error): a 10% miss is equally bad on a short or a long load. **WAPE** (total absolute error over total actual rate) and **MAE** (dollars) are reported alongside, because MAPE gives cheap loads the same weight as expensive ones and penalizes over-prediction more than under-prediction.

2.3 Leakage control

- Cleaning applied to the raw files is deterministic and row by row (date parsing; negative weights, which are sign errors, made positive). Nothing in it is learned from the data.
- Everything learned from data (imputation values, encoders, the corrupted-label filter) is part of each model and is refit inside every fold on that fold's training months only.
- Features use information available at prediction time only; the time trend is learned from training months and held at the last training day when predicting later dates.

2.4 Corrupted labels: removed from training, kept in evaluation

About 1.3% of posted rates sit in separate clusters at roughly 0.2-0.6x and 2.2-5x the usual rate per mile of comparable loads (same equipment and distance band), with empty gaps in between: for example, a 230-mile Cincinnati to St. Louis Dry Van load posted at $102 and another at $2,058 on a lane that normally pays about $590. Inside each fold, the filter learns the typical rate per mile from the training months and drops implausible *training* rows. Validation and test rows are **never** removed, because the real scoring data will contain such labels too. MAPE on plausible labels is reported separately to show how much of the error they cause.

2.5 Selection rule

Every choice (model type, trend handling, trimming, hyperparameters) was made on the cross-validation folds with one rule: **lowest mean MAPE plus one standard deviation**, so a setting must be both accurate and stable across months to win.

3. Results

3.1 Cross-validation (January to September, 4 folds)

Model	MAPE (mean ± std)	MAPE, plausible labels	WAPE	MAE
Baseline (median $/mile by equipment)	10.31% ± 0.49	8.28%	9.28%	$224
Ridge regression	5.29% ± 1.22	3.19%	5.53%	$134
LightGBM	5.42% ± 1.62	3.34%	5.43%	$131
Final hybrid (Ridge + LightGBM)	**4.15% ± 0.32**	**1.99%**	**4.30%**	**$104**

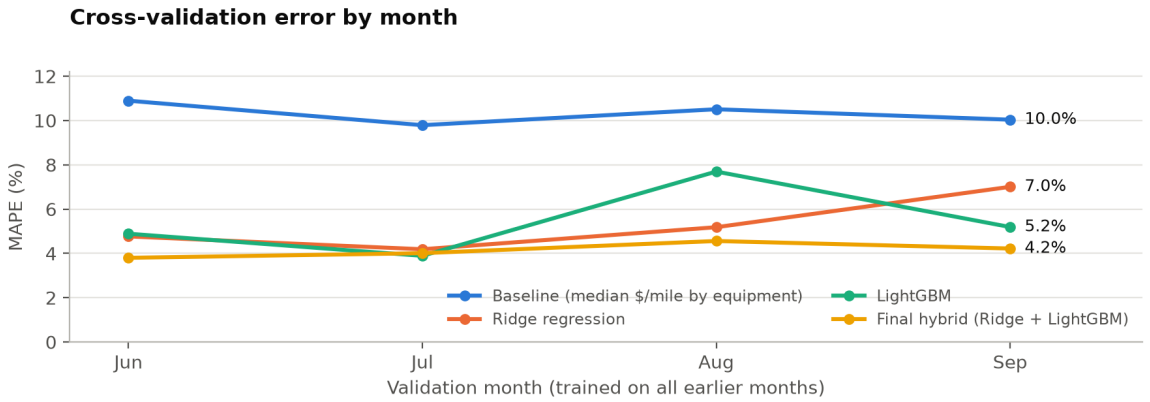


Figure 2. MAPE per validation month. Ridge and LightGBM both drifted low in later months: rates rose about 0.75% per month independently of the market index. The final model adds a time trend (held at the last training day) and combines both models, which removes the drift and halves the spread.

The final model cuts the baseline's error by 60%. Its median prediction is within about 1.6% of the actual rate in every fold. On plausible labels its error is about 2%; the corrupted labels account for the rest.

3.2 October test (trained on January to September, evaluated once)

Model	MAPE	MAPE, plausible labels	WAPE	MAE
Baseline (median $/mile by equipment)	10.93%	8.10%	9.72%	$231
Ridge regression	6.29%	3.41%	5.81%	$138
LightGBM	5.61%	2.71%	5.25%	$125
Final hybrid (Ridge + LightGBM)	**4.54%**	**1.56%**	**4.04%**	**$96**

The ranking of all models matches cross-validation, and the final model's error is less than half the baseline's (10.93%). Its 4.54% is slightly above its cross-validated 4.15%, but on plausible labels it is better (1.56% vs 1.99%): October's loads with suspect labels carry an average error of 182% and alone add about 3 points of MAPE. The median prediction is 0.8% below the actual rate: the model is not biased.

4. December prediction chart

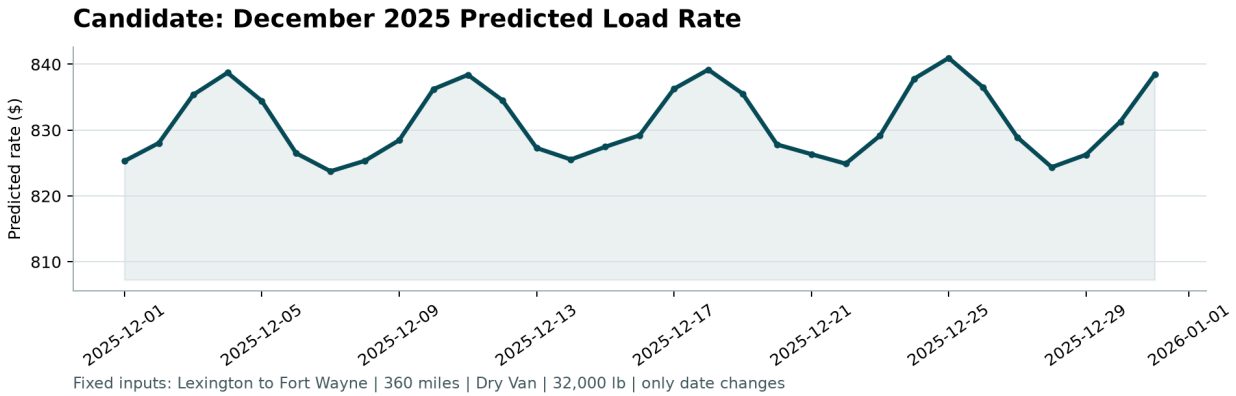


Figure 3. Output of score.py for the fixed December load (final model refit on January to October).

- **Level:** $824 to $841 for Lexington to Fort Wayne, Dry Van, 360 miles, 32,000 lb. Dry Van loads on this lane cost a median of $807 earlier in the year. The prediction sits slightly above, consistent with the upward trend held at its October level.
- **Shape:** a weekly cycle of about 1%, peaking on Thursdays and lowest on Sundays and Mondays. It follows the market index supplied for each December date, which has the same weekly cycle as in training, where rates follow it as well.
- **Limitation:** no holiday period appears in the training data, so the model treats 25 December like any other Thursday (it is the month's highest point, $841). Holiday effects are not modeled.

Reproducing the results

From the repository root, with Poetry:

```
poetry install --no-root --with report
poetry run python -m src.data_preparation   # exploration and cleaning
poetry run python -m src.train              # model comparison (cross-validation)
poetry run python -m src.tuning             # tuning experiments
poetry run python -m src.test_evaluation    # one-time October test
poetry run python -m src.predict            # submission files
poetry run python score.py --predictions validation_predictions.csv \
    --december-predictions december-chart-inputs.csv
poetry run python -m src.report             # this report
```

Each step is documented in `src/steps/`.